

Q 2.3 Given two matrices, $\bar{A} \in \mathbb{R}^{m \times n}$ and $\bar{B} \in \mathbb{R}^{m \times n}$, prove that
 $\text{tr}(\bar{A}^T \bar{B}) = \text{tr}(\bar{A} \bar{B}^T) = \text{tr}(\bar{B} \bar{A}^T) = \text{tr}(\bar{B}^T \bar{A}) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}$,

where a_{ij} and b_{ij} denote an element in the matrices $\bar{A}$ and $\bar{B}$, respectively.

$$\bar{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n} \quad \bar{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{bmatrix}_{m \times n}$$

$$\bar{A}^T = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}_{n \times m} \quad \bar{B}^T = \begin{bmatrix} b_{11} & b_{21} & \dots & b_{m1} \\ b_{12} & b_{22} & \dots & b_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1n} & b_{2n} & \dots & b_{mn} \end{bmatrix}_{n \times m}$$

$$\begin{aligned} \text{i) } \text{tr}(\bar{A}^T \bar{B}) &= a_{11}b_{11} + a_{21}b_{21} + \dots + a_{m1}b_{m1} \\ &\quad + a_{12}b_{12} + a_{22}b_{22} + \dots + a_{m2}b_{m2} \\ &\quad + \dots + a_{1n}b_{1n} + a_{2n}b_{2n} + \dots + a_{mn}b_{mn} \\ &= \sum_{i=1}^m a_{i1}b_{i1} + \sum_{i=1}^m a_{i2}b_{i2} + \dots + \sum_{i=1}^m a_{in}b_{in} \\ &= \sum_{i=1}^m [a_{i1}b_{i1} + a_{i2}b_{i2} + \dots + a_{in}b_{in}] \\ &= \sum_{i=1}^m \left[\sum_{j=1}^n a_{ij}b_{ij} \right] \end{aligned}$$

- (12)

$$+ a_{m1} b_{m1} + a_{m2} b_{m2} + \dots + a_{mn} b_{mn}$$

$$+ a_{1n}b_{1n} + a_{2n}b_{2n} + \dots + a_{mn}b_{mn}$$

$$\sum_{i=1}^m a_{i1} b_{i1} + \sum_{i=1}^m a_{i2} b_{i2} + \dots + \sum_{i=1}^m a_{in} b_{in}$$

— (2)

from (1) & (2) $\text{tr}(\bar{A}^T \bar{B}) = \text{tr}(\bar{A} \bar{B}^T)$ — (3)

Now, $\text{tr}(\bar{X}^T) = \text{tr}(\bar{X})$

$$\therefore \text{tr}((\bar{A}^T \bar{B})^T) = \text{tr}(\bar{A}^T \bar{B})$$

$$= \text{tr}(\bar{A} \bar{B}^T) \quad (\text{From (3)})$$

$$\therefore \text{tr}(\bar{B}^T \bar{A}) = \text{tr}(\bar{A} \bar{B}^T) \quad \text{--- (4)}$$

11) $\text{tr}((\bar{A} \bar{B}^T)^T) = \text{tr}(\bar{A} \bar{B}^T)$

$$\therefore \text{tr}(\bar{B} \bar{A}^T) = \text{tr}(\bar{A} \bar{B}^T) \quad \text{--- (5)}$$

From (3), (4), and (5)

$$\text{tr}(\bar{A}^T \bar{B}) = \text{tr}(\bar{A} \bar{B}^T) = \text{tr}(\bar{B}^T \bar{A}) = \text{tr}(\bar{B} \bar{A}^T) \quad - (6)$$

Hence Proved.

Q 2.2 For any two square matrices, $\bar{X} \in \mathbb{R}^{n \times n}$ and $\bar{Y} \in \mathbb{R}^{n \times n}$, show that

a. $\text{tr}(\bar{X}\bar{Y}) = \text{tr}(\bar{Y}\bar{X})$, and

b. $\text{tr}(\bar{X}^{-1}\bar{Y}\bar{X}) = \text{tr}(\bar{Y})$ if $\bar{X}$ is invertible

$$\bar{X} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix}_{n \times n}$$

$$\bar{Y} = \begin{bmatrix} y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & y_{22} & \dots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \dots & y_{nn} \end{bmatrix}_{n \times n}$$

b. Given: $\bar{X}$ is invertible. , let $\bar{Y}\bar{X} = \bar{A}$

$$\begin{aligned} \Rightarrow & \text{tr}(\bar{X}^{-1}\bar{Y}\bar{X}) \\ &= \text{tr}(\bar{X}^{-1}\bar{A}) \\ &= \text{tr}(\bar{A}\bar{X}^{-1}) \quad (\because \text{tr}(XY) = \text{tr}(YX)) \\ &= \text{tr}(\bar{Y}\bar{X}\bar{X}^{-1}) \\ &= \text{tr}(\bar{Y}\bar{I}) \\ &= \text{tr}(\bar{Y}) \end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$

Hence Proved.

Q 2.3 Given two sets of m vectors, $\bar{x}_i \in \mathbb{R}^n$ and $\bar{y}_i \in \mathbb{R}^n$ for all $i=1, 2, \dots, m$, verify that the summations $\sum_{i=1}^m \bar{x}_i \bar{x}_i^T$ and $\sum_{i=1}^m \bar{x}_i \bar{y}_i^T$ can be vectorized as the following matrix multiplication:

$$\sum_{i=1}^m \bar{x}_i \bar{x}_i^T = \bar{X}\bar{X}^T \text{ and } \sum_{i=1}^m \bar{x}_i \bar{y}_i^T = \bar{X}\bar{Y}^T,$$

where $\bar{X} = [\bar{x}_1 \bar{x}_2 \dots \bar{x}_m] \in \mathbb{R}^{n \times m}$ and $\bar{Y} = [\bar{y}_1 \bar{y}_2 \dots \bar{y}_m] \in \mathbb{R}^{n \times m}$.

$$\bar{x}^T = \begin{bmatrix} \bar{x}_1^T \\ \bar{x}_2^T \\ \vdots \\ \bar{x}_m^T \end{bmatrix}$$

$$\bar{x}\bar{x}^T = [\bar{x}_1 \ \bar{x}_2 \ \dots \ \bar{x}_m] \begin{bmatrix} \bar{x}_1^T \\ \bar{x}_2^T \\ \vdots \\ \bar{x}_m^T \end{bmatrix}$$

$$= \bar{x}_1\bar{x}_1^T + \bar{x}_2\bar{x}_2^T + \dots + \bar{x}_m\bar{x}_m^T$$

$$= \sum_{i=1}^m \bar{x}_i\bar{x}_i^T$$

Now, $\bar{y} = [\bar{y}_1 \ \bar{y}_2 \ \dots \ \bar{y}_m]$

$$\bar{y}^T = \begin{bmatrix} \bar{y}_1^T \\ \bar{y}_2^T \\ \bar{y}_3^T \\ \vdots \\ \bar{y}_m^T \end{bmatrix}$$

$$\therefore \bar{x}\bar{y}^T = [\bar{x}_1 \ \bar{x}_2 \ \dots \ \bar{x}_m] \begin{bmatrix} \bar{y}_1^T \\ \bar{y}_2^T \\ \vdots \\ \bar{y}_m^T \end{bmatrix}$$

$$= \bar{x}_1\bar{y}_1^T + \bar{x}_2\bar{y}_2^T + \dots + \bar{x}_m\bar{y}_m^T$$

$$= \sum_{i=1}^m \bar{x}_i\bar{y}_i^T$$

Hence, proved.

Q2.4 Given $\bar{x} \in \mathbb{R}^n$, $\bar{z} \in \mathbb{R}^m$, and $\bar{A} \in \mathbb{R}^{m \times n}$ ($m < n$),

a. prove that $\bar{z}^T \bar{A} \bar{x} = \text{tr}(\bar{x} \bar{z}^T \bar{A})$,

b. compute the derivative $(\partial/\partial \bar{x}) \|\bar{z} - \bar{A} \bar{x}\|^2$, and

c. compute the derivative $(\partial/\partial \bar{A}) \|\bar{z} - \bar{A} \bar{x}\|^2$.

$$a. \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \quad \bar{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{bmatrix}_{m \times 1} \quad \bar{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

$$\bar{z}^T = [z_1 \ z_2 \ \dots \ z_m]_{1 \times m}$$

$$\therefore \bar{z}^T \bar{A} = \begin{bmatrix} z_1 a_{11} + z_2 a_{21} + \dots + z_m a_{m1} \\ z_1 a_{12} + z_2 a_{22} + \dots + z_m a_{m2} \\ \vdots \\ z_1 a_{1n} + z_2 a_{2n} + \dots + z_m a_{mn} \end{bmatrix}_{1 \times n}$$

$$\begin{aligned} (\bar{z}^T \bar{A}) \bar{x} &= x_1 (z_1 a_{11} + z_2 a_{21} + \dots + z_m a_{m1}) \\ &\quad + x_2 (z_1 a_{12} + z_2 a_{22} + \dots + z_m a_{m2}) \\ &\quad + \dots + x_n (z_1 a_{1n} + z_2 a_{2n} + \dots + z_m a_{mn}) \\ &= \text{L.H.S.} \end{aligned}$$

$$\bar{x} (\bar{z}^T \bar{A}) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \begin{bmatrix} z_1 a_{11} + z_2 a_{21} + \dots + z_m a_{m1} \\ z_1 a_{12} + z_2 a_{22} + \dots + z_m a_{m2} \\ \vdots \\ z_1 a_{1n} + z_2 a_{2n} + \dots + z_m a_{mn} \end{bmatrix}_{1 \times n}$$

$$= \begin{bmatrix} x_1(z_1q_{11} + z_2q_{21} + \dots + z_mq_{m1}) & x_1(z_1q_{12} + \dots + z_mq_{m2}) & x_1(z_1q_{1n} + \dots + z_mq_{mn}) \\ x_2(z_1q_{11} + z_2q_{21} + \dots + z_mq_{m1}) & x_2(z_1q_{12} + \dots + z_mq_{m2}) & x_2(z_1q_{1n} + \dots + z_mq_{mn}) \\ \vdots & \vdots & \vdots \\ x_n(z_1q_{11} + z_2q_{21} + \dots + z_mq_{m1}) & x_n(z_1q_{12} + \dots + z_mq_{m2}) & x_n(z_1q_{1n} + \dots + z_mq_{mn}) \end{bmatrix}$$

RHS

$$\therefore \text{tr}(\bar{x} \bar{z}^T \bar{A}) = x_1(z_1q_{11} + z_2q_{21} + \dots + z_mq_{m1}) \\ + x_2(z_1q_{12} + \dots + z_mq_{m2}) \\ + \dots + x_n(z_1q_{1n} + \dots + z_mq_{mn})$$

$$= \text{R.H.S.}$$

Hence proved

b. $\frac{\partial}{\partial \bar{x}} \|\bar{z} - \bar{A}\bar{x}\|^2$

$$\|\bar{z} - \bar{A}\bar{x}\|^2 = (\bar{z} - \bar{A}\bar{x})^T (\bar{z} - \bar{A}\bar{x}) \\ = (\bar{z}^T - \bar{x}^T \bar{A}^T) (\bar{z} - \bar{A}\bar{x}) \\ = \bar{z}^T \bar{z} - \bar{z}^T \bar{A}\bar{x} - \bar{x}^T \bar{A}^T \bar{z} + \bar{x}^T \bar{A}^T \bar{A} \bar{x}$$

Now, $\bar{z}^T \bar{A}\bar{x}$ and $\bar{x}^T \bar{A}^T \bar{z}$ both are scalar quantities

$$\therefore (\bar{z}^T \bar{A}\bar{x}) = (\bar{z}^T \bar{A}\bar{x})^T \\ = (\bar{A}\bar{x})^T \bar{z} = \bar{x}^T \bar{A}^T \bar{z}$$

$$\therefore \|\bar{z} - \bar{A}\bar{x}\|^2 = \bar{z}^T \bar{z} - 2\bar{z}^T \bar{A}\bar{x} + \bar{x}^T \bar{A}^T \bar{A} \bar{x}$$

$$\text{OR } \|\bar{z} - \bar{A}\bar{x}\|^2 = \bar{z}^T \bar{z} - 2\bar{x}^T \bar{A}^T \bar{z} + \bar{x}^T \bar{A}^T \bar{A} \bar{x}$$

$$\frac{\partial}{\partial \bar{x}} (\|\bar{z} - \bar{A}\bar{x}\|^2) = \frac{\partial}{\partial \bar{x}} (\bar{z}^T \bar{z}) - 2 \frac{\partial}{\partial \bar{x}} (\bar{x}^T \bar{A}^T \bar{z}) + \frac{\partial}{\partial \bar{x}} (\bar{x}^T \bar{B} \bar{x})$$

[where $\bar{B} = \bar{A}^T \bar{A}$ is always symmetric matrix]

$$\therefore \frac{\partial}{\partial \bar{x}} \|\bar{z} - \bar{A}\bar{x}\|^2 = 0 - 2\bar{A}^T \bar{z} + 2\bar{B}\bar{x} \\ = -2\bar{A}^T \bar{z} + 2\bar{A}^T \bar{A} \bar{x}$$

c. $\frac{\partial}{\partial \bar{A}} \|\bar{z} - \bar{A}\bar{x}\|^2 = \frac{\partial}{\partial \bar{A}} (\bar{z}^T \bar{z}) - 2 \frac{\partial}{\partial \bar{A}} (\bar{z}^T \bar{A} \bar{x}) + \frac{\partial}{\partial \bar{A}} (\bar{x}^T \bar{B} \bar{x})$

$$= 0 - 2\bar{z} \bar{x}^T + \frac{\partial (\bar{B})}{\partial \bar{A}} \frac{\partial}{\partial \bar{B}} (\bar{x}^T \bar{B} \bar{x})$$

$$= -2\bar{z} \bar{x}^T + 2\bar{A} \bar{x} \bar{x}^T$$